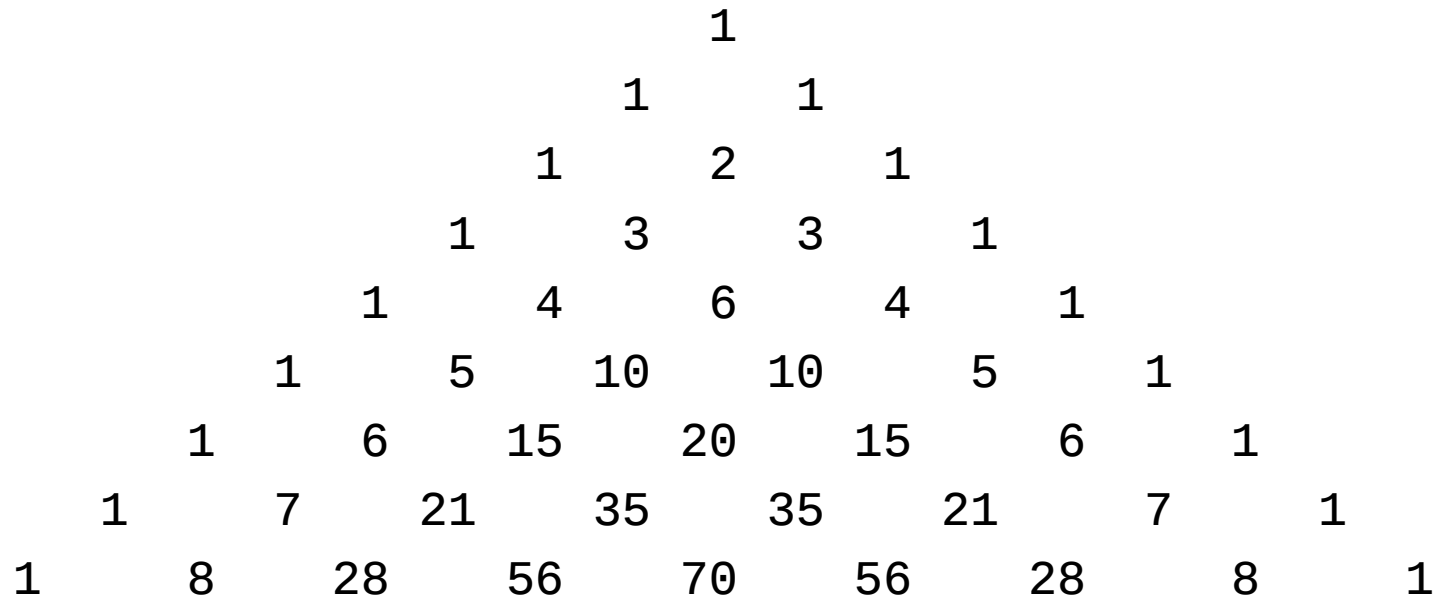
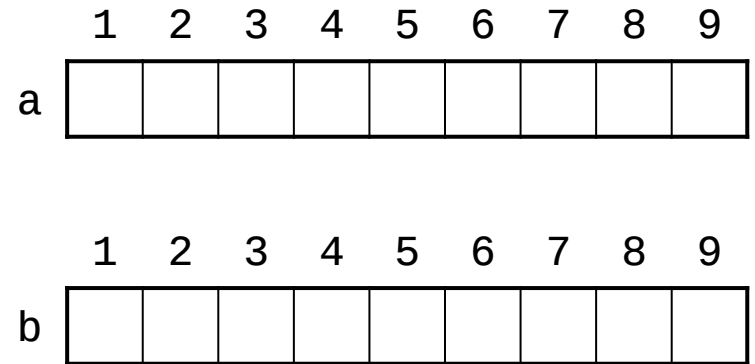
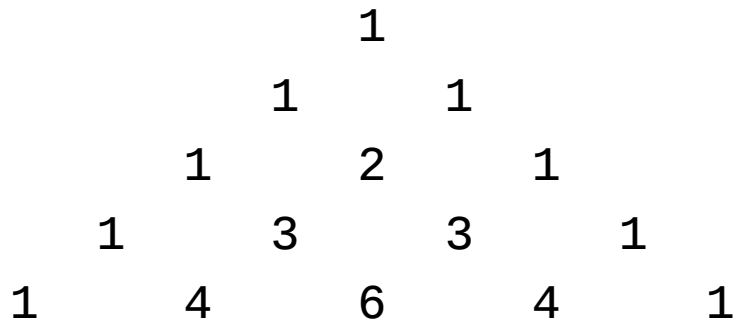


Pascal triangle

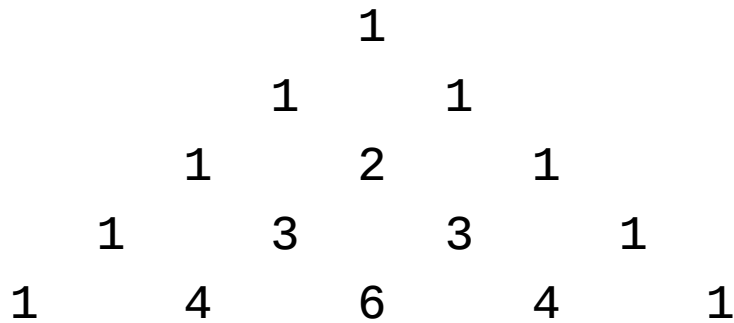
How many levels? (1-13): 9



Pascal triangle - two arrays



Pascal triangle - two arrays

[illegible]

Pascal triangle - two arrays

				1					
			1		1				
		1		2		1			
	1		3		3		1		
1		4		6		4		1	

	1	2	3	4	5	6	7	8	9
a	1								

	1	2	3	4	5	6	7	8	9
b	1	1							

Pascal triangle - two arrays

				1					
			1		1				
		1		2		1			
	1		3		3		1		
1		4		6		4		1	

	1	2	3	4	5	6	7	8	9
a	1	1							

	1	2	3	4	5	6	7	8	9
b	1	1							

Pascal triangle - two arrays

				1					
			1		1				
		1		2		1			
	1		3		3		1		
1		4		6		4		1	

	1	2	3	4	5	6	7	8	9
a	1	1							

	1	2	3	4	5	6	7	8	9
b	1	2	1						

Pascal triangle - two arrays

				1					
			1		1				
		1		2		1			
	1		3		3		1		
1		4		6		4		1	

	1	2	3	4	5	6	7	8	9
a	1	2	1						

	1	2	3	4	5	6	7	8	9
b	1	2	1						

Pascal triangle - two arrays

				1					
			1		1				
		1		2		1			
	1		3		3		1		
1		4		6		4		1	

	1	2	3	4	5	6	7	8	9
a	1	2	1						

	1	2	3	4	5	6	7	8	9
b	1	3	3	1					

Pascal triangle - two arrays

				1					
			1		1				
		1		2		1			
	1		3		3		1		
1		4		6		4		1	

	1	2	3	4	5	6	7	8	9
a	1	3	3	1					

	1	2	3	4	5	6	7	8	9
b	1	3	3	1					

	1	2	3	4	5	6	7	8	9
a	1								

	1	2	3	4	5	6	7	8	9
b	1	1							

`a[1] = 1;`

	1	2	3	4	5	6	7	8	9
a	1								

	1	2	3	4	5	6	7	8	9
b	1	1							

`a[1] = 1;`

	1	2	3	4	5	6	7	8	9
a	1	1							

	1	2	3	4	5	6	7	8	9
b	1	1							

```
a[1] = 1;
```

```
b[1] = a[1];
```

```
b[2] = 1;
```

```
a[1] = b[1];
```

```
a[2] = b[2];
```

	1	2	3	4	5	6	7	8	9
a	1	1							

	1	2	3	4	5	6	7	8	9
b	1	1							

```
a[1] = 1;
```

```
b[1] = a[1];
```

```
b[2] = 1;
```

```
a[1] = b[1];
```

```
a[2] = b[2];
```

	1	2	3	4	5	6	7	8	9
a	1	1							

	1	2	3	4	5	6	7	8	9
b	1	2	1						

```
a[1] = 1;
```

```
b[1] = a[1];
```

```
b[2] = 1;
```

```
a[1] = b[1];
```

```
a[2] = b[2];
```

	1	2	3	4	5	6	7	8	9
a	1	2	1						

	1	2	3	4	5	6	7	8	9
b	1	2	1						

```
a[1] = 1;
```

```
b[1] = a[1];
```

```
b[2] = 1;
```

```
a[1] = b[1];
```

```
a[2] = b[2];
```

```
b[1] = a[1];
```

```
b[2] = a[1] + a[2];
```

```
b[3] = 1;
```

```
a[1] = b[1];
```

```
a[2] = b[2];
```

```
a[3] = b[3];
```

	1	2	3	4	5	6	7	8	9
a	1	2	1						

	1	2	3	4	5	6	7	8	9
b	1	2	1						

```
a[1] = 1;
```

```
b[1] = a[1];
```

```
b[2] = 1;
```

```
a[1] = b[1];
```

```
a[2] = b[2];
```

```
b[1] = a[1];
```

```
b[2] = a[1] + a[2];
```

```
b[3] = 1;
```

```
a[1] = b[1];
```

```
a[2] = b[2];
```

```
a[3] = b[3];
```

	1	2	3	4	5	6	7	8	9
a	1	2	1						

	1	2	3	4	5	6	7	8	9
b	1	3	3	1					

```
a[1] = 1;
```

```
b[1] = a[1];
```

```
b[2] = 1;
```

```
a[1] = b[1];
```

```
a[2] = b[2];
```

```
b[1] = a[1];
```

```
b[2] = a[1] + a[2];
```

```
b[3] = 1;
```

```
a[1] = b[1];
```

```
a[2] = b[2];
```

```
a[3] = b[3];
```

	1	2	3	4	5	6	7	8	9
a	1	3	3	1					

	1	2	3	4	5	6	7	8	9
b	1	3	3	1					

```
a[1] = 1;
```

```
b[1] = a[1];
```

```
b[2] = 1;
```

```
a[1] = b[1];
```

```
a[2] = b[2];
```

```
b[1] = a[1];
```

```
b[2] = a[1] + a[2];
```

```
b[3] = 1;
```

```
a[1] = b[1];
```

```
a[2] = b[2];
```

```
a[3] = b[3];
```

```
b[1] = a[1];
```

```
b[2] = a[1] + a[2];
```

```
b[3] = a[2] + a[3];
```

```
b[4] = 1;
```

```
a[1] = b[1];
```

```
a[2] = b[2];
```

```
a[3] = b[3];
```

```
a[4] = b[4];
```

	1	2	3	4	5	6	7	8	9
a	1	3	3	1					

	1	2	3	4	5	6	7	8	9
b	1	3	3	1					

```
a[1] = 1;
```

```
b[1] = a[1];
```

```
b[2] = 1;
```

```
a[1] = b[1];
```

```
a[2] = b[2];
```

```
b[1] = a[1];
```

```
b[2] = a[1] + a[2];
```

```
b[3] = 1;
```

```
a[1] = b[1];
```

```
a[2] = b[2];
```

```
a[3] = b[3];
```

```
b[1] = a[1];
```

```
b[2] = a[1] + a[2];
```

```
b[3] = a[2] + a[3];
```

```
b[4] = 1;
```

```
a[1] = b[1];
```

```
a[2] = b[2];
```

```
a[3] = b[3];
```

```
a[4] = b[4];
```

	1	2	3	4	5	6	7	8	9
a	1	3	3	1					

	1	2	3	4	5	6	7	8	9
b	1	4	6	4	1				

```
a[1] = 1;
```

```
b[1] = a[1];
```

```
b[2] = 1;
```

```
a[1] = b[1];
```

```
a[2] = b[2];
```

```
b[1] = a[1];
```

```
b[2] = a[1] + a[2];
```

```
b[3] = 1;
```

```
a[1] = b[1];
```

```
a[2] = b[2];
```

```
a[3] = b[3];
```

```
b[1] = a[1];
```

```
b[2] = a[1] + a[2];
```

```
b[3] = a[2] + a[3];
```

```
b[4] = 1;
```

```
a[1] = b[1];
```

```
a[2] = b[2];
```

```
a[3] = b[3];
```

```
a[4] = b[4];
```

	1	2	3	4	5	6	7	8	9
a	1	4	6	4	1				

	1	2	3	4	5	6	7	8	9
b	1	4	6	4	1				

```
a[1] = 1;
```

```
b[1] = a[1];
```

```
b[2] = 1;
```

```
a[1] = b[1];
```

```
a[2] = b[2];
```

```
b[1] = a[1];
```

```
b[2] = a[1] + a[2];
```

```
b[3] = 1;
```

```
a[1] = b[1];
```

```
a[2] = b[2];
```

```
a[3] = b[3];
```

```
b[1] = a[1];
```

```
b[2] = a[1] + a[2];
```

```
b[3] = a[2] + a[3];
```

```
b[4] = 1;
```

```
a[1] = b[1];
```

```
a[2] = b[2];
```

```
a[3] = b[3];
```

```
a[4] = b[4];
```

	1	2	3	4	5	6	7	8	9
a	1	4	6	4	1				

	1	2	3	4	5	6	7	8	9
b	1	4	6	4	1				

```
b[1] = a[1];
```

```
b[2] = a[1] + a[2];
```

```
b[3] = a[2] + a[3];
```

```
b[4] = a[3] + a[4];
```

```
b[5] = 1;
```

```
a[1] = b[1];
```

```
a[2] = b[2];
```

```
a[3] = b[3];
```

```
a[4] = b[4];
```

```
a[5] = b[5];
```

```
a[1] = 1;
```

```
a[1] = b[1];  
a[2] = b[2];
```

```
a[1] = b[1];  
a[2] = b[2];  
a[3] = b[3];
```

```
a[1] = b[1];  
a[2] = b[2];  
a[3] = b[3];  
a[4] = b[4];
```

```
a[1] = b[1];  
a[2] = b[2];  
a[3] = b[3];  
a[4] = b[4];  
a[5] = b[5];
```

```
a[1] = 1;
```

```
a[1] = b[1];  
a[2] = b[2];
```

```
a[1] = b[1];  
a[2] = b[2];  
a[3] = b[3];
```

```
a[1] = b[1];  
a[2] = b[2];  
a[3] = b[3];  
a[4] = b[4];
```

```
a[1] = b[1];  
a[2] = b[2];  
a[3] = b[3];  
a[4] = b[4];  
a[5] = b[5];
```

```
for( int k = 1; k <= 2; k++ )  
    a[k] = b[k];
```

```
for( int k = 1; k <= 3; k++ )  
    a[k] = b[k];
```

```
for( int k = 1; k <= 4; k++ )  
    a[k] = b[k];
```

```
for( int k = 1; k <= 5; k++ )  
    a[k] = b[k];
```

```
for( int i = 2; i < 6; i++ )  
{  
    for( int k = 1; k <= i; k++ )  
        a[k] = b[k];  
}
```

```
for( int k = 1; k <= 2; k++ )  
    a[k] = b[k];
```

```
for( int k = 1; k <= 3; k++ )  
    a[k] = b[k];
```

```
for( int k = 1; k <= 4; k++ )  
    a[k] = b[k];
```

```
for( int k = 1; k <= 5; k++ )  
    a[k] = b[k];
```

```
a[1] = 1;
```

```
b[1] = a[1];  
b[2] = 1;
```

```
b[1] = a[1];  
b[2] = a[1] + a[2];  
b[3] = 1;
```

```
b[1] = a[1];  
b[2] = a[1] + a[2];  
b[3] = a[2] + a[3];  
b[4] = 1;
```

```
b[1] = a[1];  
b[2] = a[1] + a[2];  
b[3] = a[2] + a[3];  
b[4] = a[3] + a[4];  
b[5] = 1;
```

```
a[1] = 1;
```

```
b[1] = a[1];  
b[2] = 1;
```

```
b[1] = a[1];  
b[2] = a[1] + a[2];  
b[3] = 1;
```

```
b[1] = a[1];  
b[2] = a[1] + a[2];  
b[3] = a[2] + a[3];  
b[4] = 1;
```

```
b[1] = a[1];  
b[2] = a[1] + a[2];  
b[3] = a[2] + a[3];  
b[4] = a[3] + a[4];  
b[5] = 1;
```

```
b[1] = a[1];  
for( int j = 2; j <= 1; j++ )  
    b[j] = a[j-1] + a[j];  
b[i+1] = 1;
```

```
b[1] = a[1];  
for( int j = 2; j <= 2; j++ )  
    b[j] = a[j-1] + a[j];  
b[i+1] = 1;
```

```
b[1] = a[1];  
for( int j = 2; j <= 3; j++ )  
    b[j] = a[j-1] + a[j];  
b[i+1] = 1;
```

```
b[1] = a[1];  
for( int j = 2; j <= 4; j++ )  
    b[j] = a[j-1] + a[j];  
b[i+1] = 1;
```

```

for( int i = 1; i < 5; i++ )
{
    b[1] = a[1];
    for( int j = 2; j <= i; j++ )
        b[j] = a[j-1] + a[j];
    b[i+1] = 1;
}

```

```

b[1] = a[1];
for( int j = 2; j <= 1; j++ )
    b[j] = a[j-1] + a[j];
b[i+1] = 1;

```

```

b[1] = a[1];
for( int j = 2; j <= 2; j++ )
    b[j] = a[j-1] + a[j];
b[i+1] = 1;

```

```

b[1] = a[1];
for( int j = 2; j <= 3; j++ )
    b[j] = a[j-1] + a[j];
b[i+1] = 1;

```

```

b[1] = a[1];
for( int j = 2; j <= 4; j++ )
    b[j] = a[j-1] + a[j];
b[i+1] = 1;

```

```
for( int i = 2; i < 6; i++ )  
{  
    for( int k = 1; k <= i; k++ )  
        a[k] = b[k];  
}
```

```
for( int i = 1; i < 5; i++ )  
{  
    b[1] = a[1];  
    for( int j = 2; j <= i; j++ )  
        b[j] = a[j-1] + a[j];  
    b[i+1] = 1;  
}
```

```
for( int i = 2; i < 6; i++ )  
{  
    for( int k = 1; k <= i; k++ )  
        a[k] = b[k];  
}
```

```
for( int i = 1; i < 5; i++ )  
{  
    b[1] = a[1];  
    for( int j = 2; j <= i; j++ )  
        b[j] = a[j-1] + a[j];  
    b[i+1] = 1;  
}
```

```
for( int i = 1; i < 5; i++ )  
{  
    for( int k = 1; k <= i+1; k++ )  
        a[k] = b[k];  
}
```

```
for( int i = 1; i < 5; i++ )  
{  
    b[1] = a[1];  
    for( int j = 2; j <= i; j++ )  
        b[j] = a[j-1] + a[j];  
    b[i+1] = 1;  
}
```

```
for( int i = 1; i < 5; i++ )
{
    b[1] = a[1];
    for( int j = 2; j <= i; j++ )
        b[j] = a[j-1] + a[j];
    b[i+1] = 1;

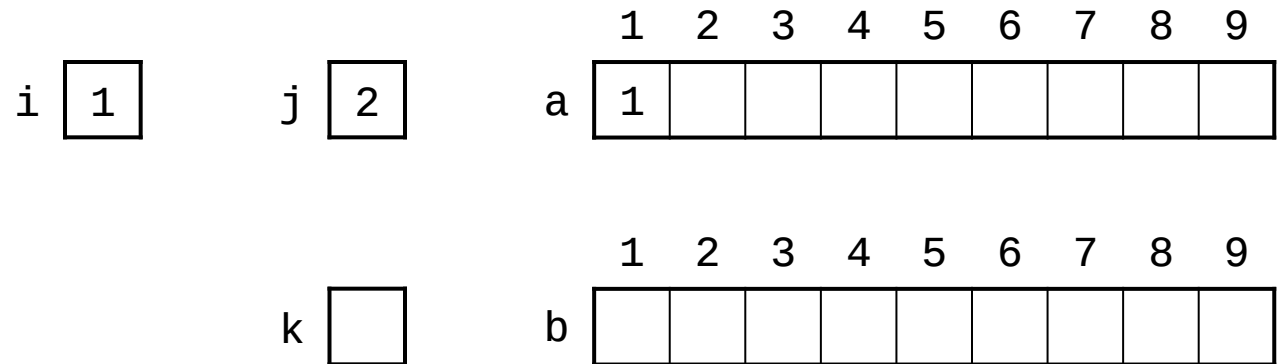
    for( int k = 1; k <= i+1; k++ )
        a[k] = b[k];
}
```

```
for( int i = 1; i < 5; i++ )
{
    for( int k = 1; k <= i+1; k++ )
        a[k] = b[k];
}

for( int i = 1; i < 5; i++ )
{
    b[1] = a[1];
    for( int j = 2; j <= i; j++ )
        b[j] = a[j-1] + a[j];
    b[i+1] = 1;
}
```

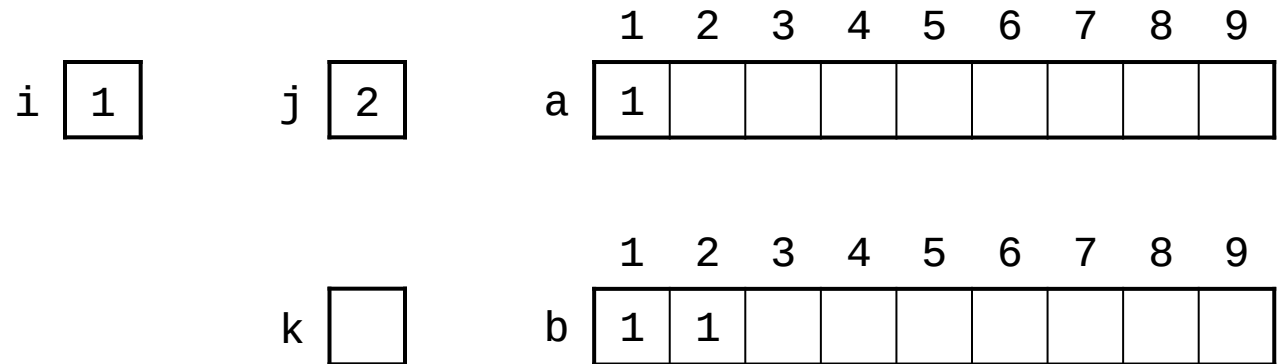
```
int main()
{
    int level, a[100], b[100];
    cout << "How many levels? (1-13): ";
    cin >> level;
    a[1] = 1;
    for( int i = 1; i < level; i++ )
    {
        b[1] = a[1];
        for( int j = 2; j <= i; j++ )
            b[j] = a[j-1] + a[j];
        b[i+1] = 1;

        for( int k = 1; k <= i+1; k++ )
            a[k] = b[k];
    }
}
```



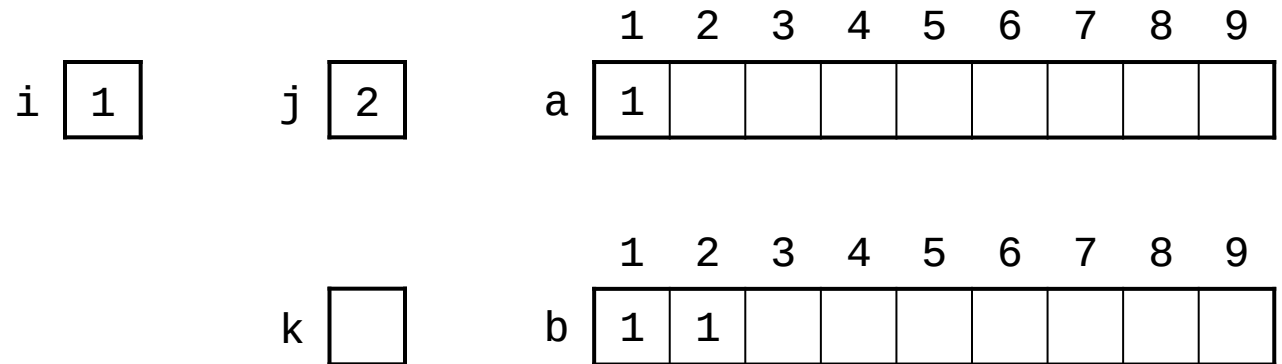
```
int main()
{
    int level, a[100], b[100];
    cout << "How many levels? (1-13): ";
    cin >> level;
    a[1] = 1;
    for( int i = 1; i < level; i++ )
    {
        b[1] = a[1];
        for( int j = 2; j <= i; j++ )
            b[j] = a[j-1] + a[j];
        b[i+1] = 1;

        for( int k = 1; k <= i+1; k++ )
            a[k] = b[k];
    }
}
```



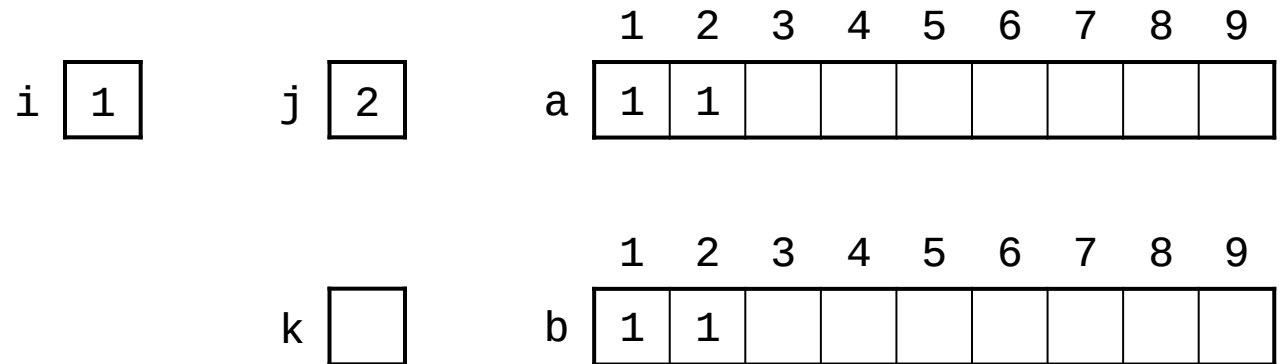
```
int main()
{
    int level, a[100], b[100];
    cout << "How many levels? (1-13): ";
    cin >> level;
    a[1] = 1;
    for( int i = 1; i < level; i++ )
    {
        b[1] = a[1];
        for( int j = 2; j <= i; j++ )
            b[j] = a[j-1] + a[j];
        b[i+1] = 1;

        for( int k = 1; k <= i+1; k++ )
            a[k] = b[k];
    }
}
```



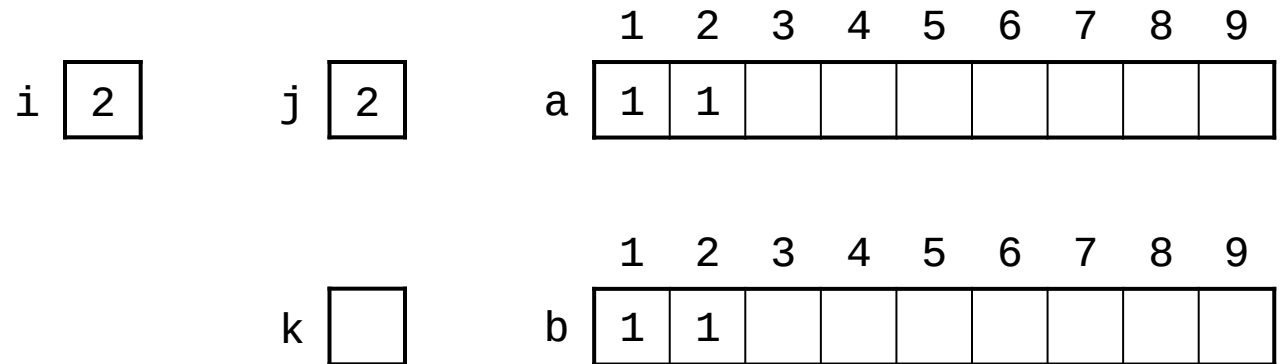
```
int main()
{
    int level, a[100], b[100];
    cout << "How many levels? (1-13): ";
    cin >> level;
    a[1] = 1;
    for( int i = 1; i < level; i++ )
    {
        b[1] = a[1];
        for( int j = 2; j <= i; j++ )
            b[j] = a[j-1] + a[j];
        b[i+1] = 1;

        for( int k = 1; k <= i+1; k++ )
            a[k] = b[k];
    }
}
```



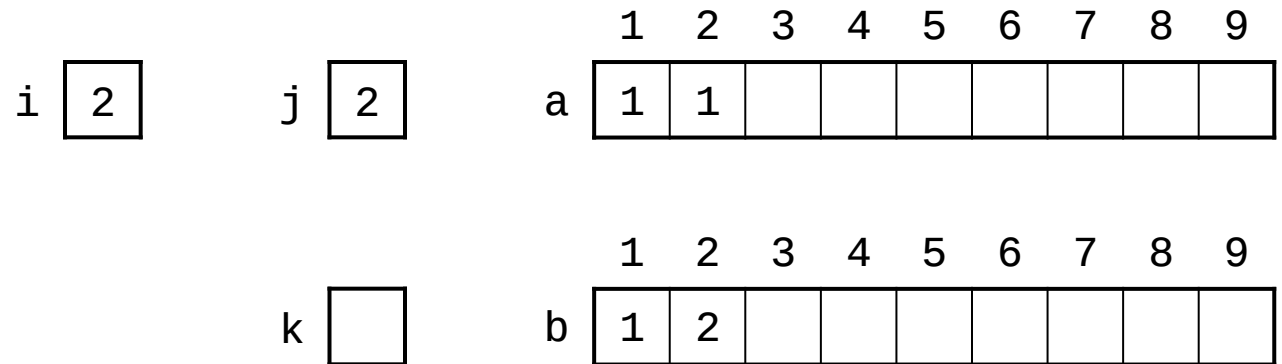
```
int main()
{
    int level, a[100], b[100];
    cout << "How many levels? (1-13): ";
    cin >> level;
    a[1] = 1;
    for( int i = 1; i < level; i++ )
    {
        b[1] = a[1];
        for( int j = 2; j <= i; j++ )
            b[j] = a[j-1] + a[j];
        b[i+1] = 1;

        for( int k = 1; k <= i+1; k++ )
            a[k] = b[k];
    }
}
```



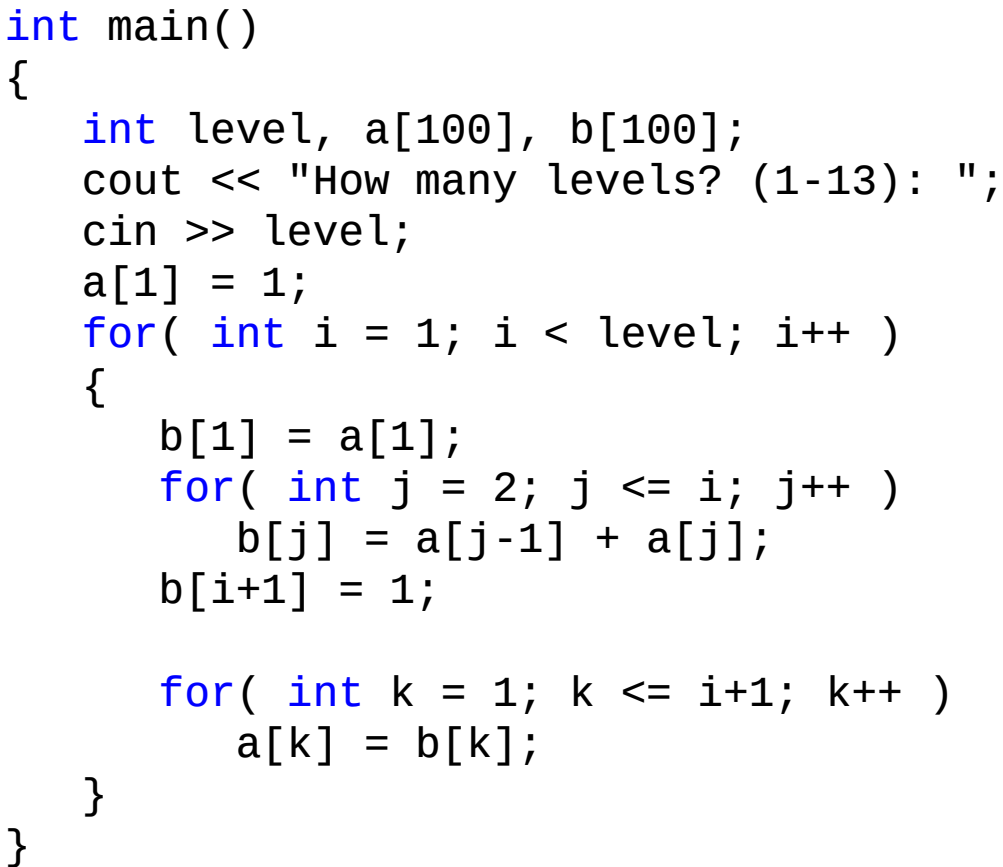
```
int main()
{
    int level, a[100], b[100];
    cout << "How many levels? (1-13): ";
    cin >> level;
    a[1] = 1;
    for( int i = 1; i < level; i++ )
    {
        b[1] = a[1];
        for( int j = 2; j <= i; j++ )
            b[j] = a[j-1] + a[j];
        b[i+1] = 1;

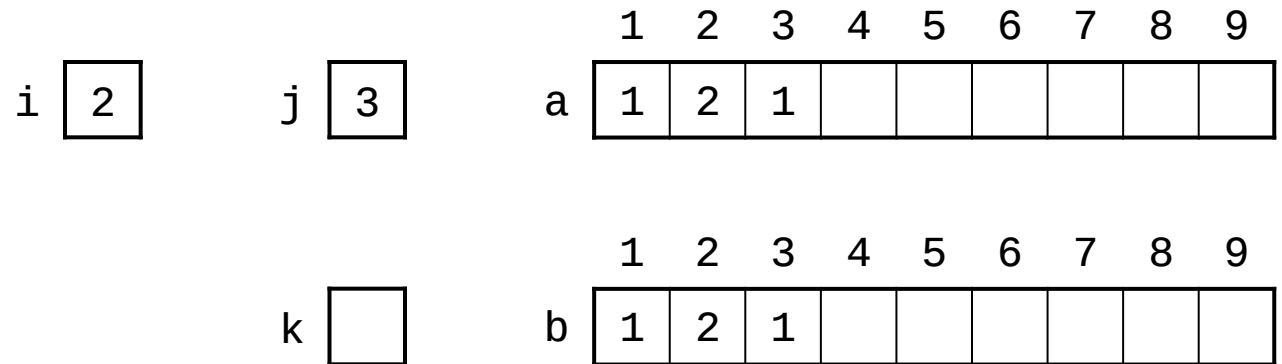
        for( int k = 1; k <= i+1; k++ )
            a[k] = b[k];
    }
}
```



```
int main()
{
    int level, a[100], b[100];
    cout << "How many levels? (1-13): ";
    cin >> level;
    a[1] = 1;
    for( int i = 1; i < level; i++ )
    {
        b[1] = a[1];
        for( int j = 2; j <= i; j++ )
            b[j] = a[j-1] + a[j];
        b[i+1] = 1;

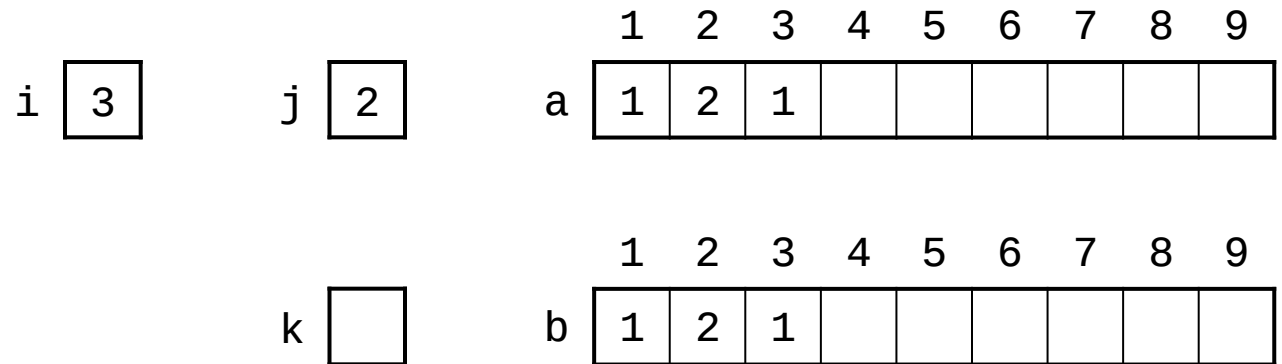
        for( int k = 1; k <= i+1; k++ )
            a[k] = b[k];
    }
}
```





```
int main()
{
    int level, a[100], b[100];
    cout << "How many levels? (1-13): ";
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            b[j] = a[j-1] + a[j];
        b[i+1] = 1;

        for( int k = 1; k <= i+1; k++ )
            a[k] = b[k];
    }
}
```

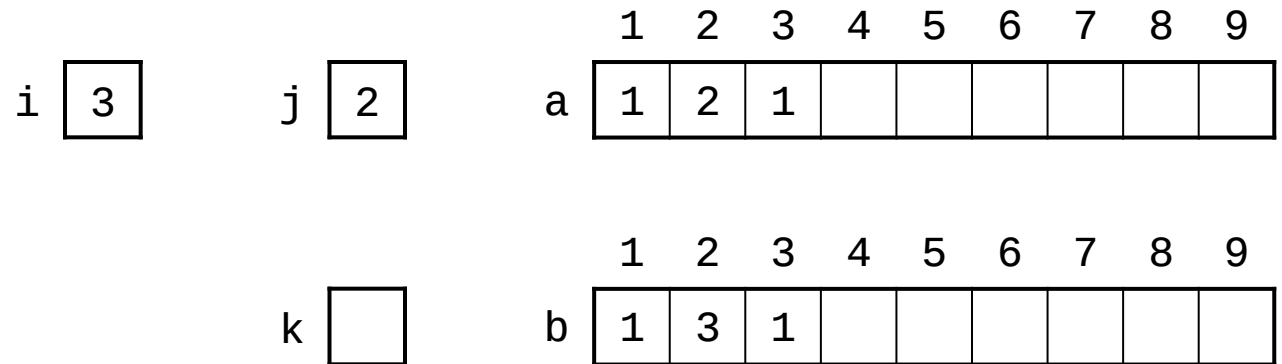


```

int main()
{
    int level, a[100], b[100];
    cout << "How many levels? (1-13): ";
    cin >> level;
    a[1] = 1;
    for( int i = 1; i < level; i++ )
    {
        b[1] = a[1];
        for( int j = 2; j <= i; j++ )
            b[j] = a[j-1] + a[j];
        b[i+1] = 1;

        for( int k = 1; k <= i+1; k++ )
            a[k] = b[k];
    }
}

```

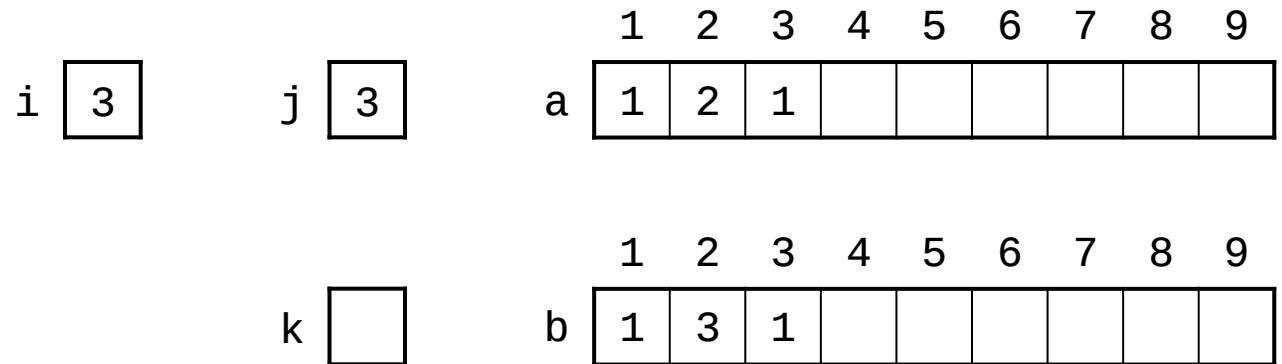


```

int main()
{
    int level, a[100], b[100];
    cout << "How many levels? (1-13): ";
    cin >> level;
    a[1] = 1;
    for( int i = 1; i < level; i++ )
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            b[j] = a[j-1] + a[j];
        b[i+1] = 1;

        for( int k = 1; k <= i+1; k++ )
            a[k] = b[k];
    }
}

```

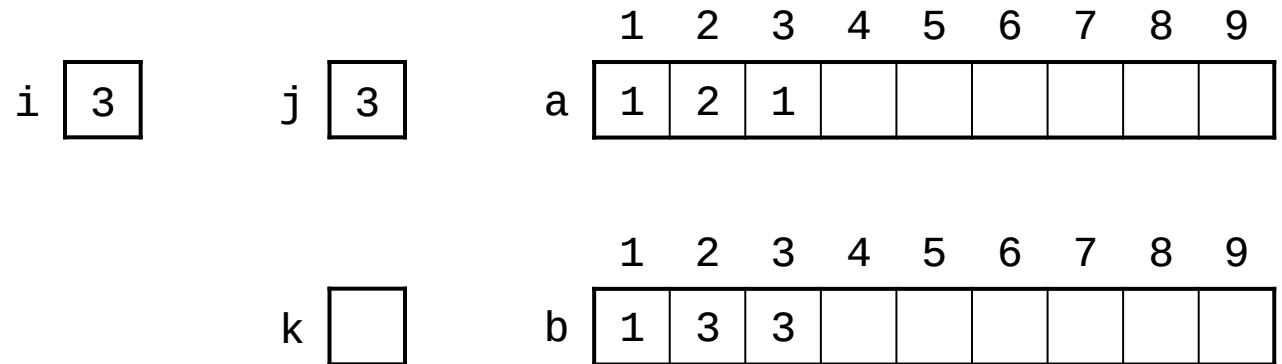


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{
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    cout << "How many levels? (1-13): ";
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    a[1] = 1;
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            b[j] = a[j-1] + a[j];
        b[i+1] = 1;

        for( int k = 1; k <= i+1; k++ )
            a[k] = b[k];
    }
}

```

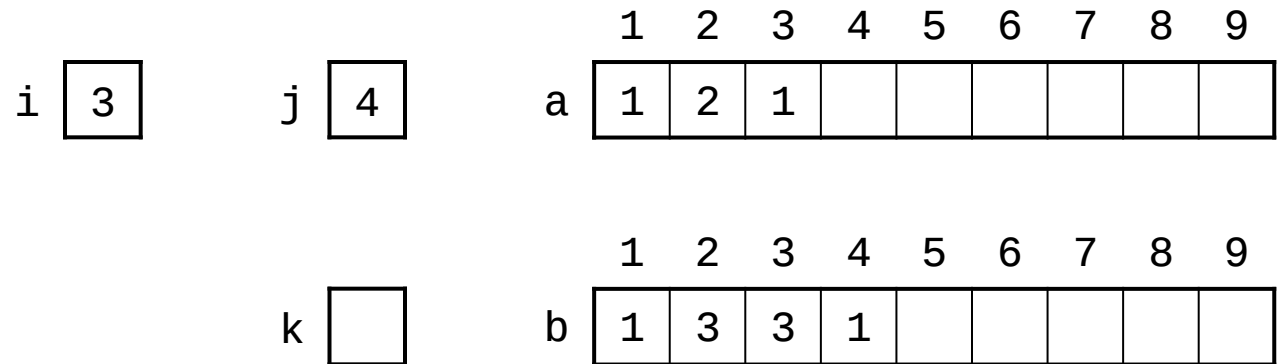


```

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{
    int level, a[100], b[100];
    cout << "How many levels? (1-13): ";
    cin >> level;
    a[1] = 1;
    for( int i = 1; i < level; i++ )
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        for( int j = 2; j <= i; j++ )
            b[j] = a[j-1] + a[j];
        b[i+1] = 1;

        for( int k = 1; k <= i+1; k++ )
            a[k] = b[k];
    }
}

```

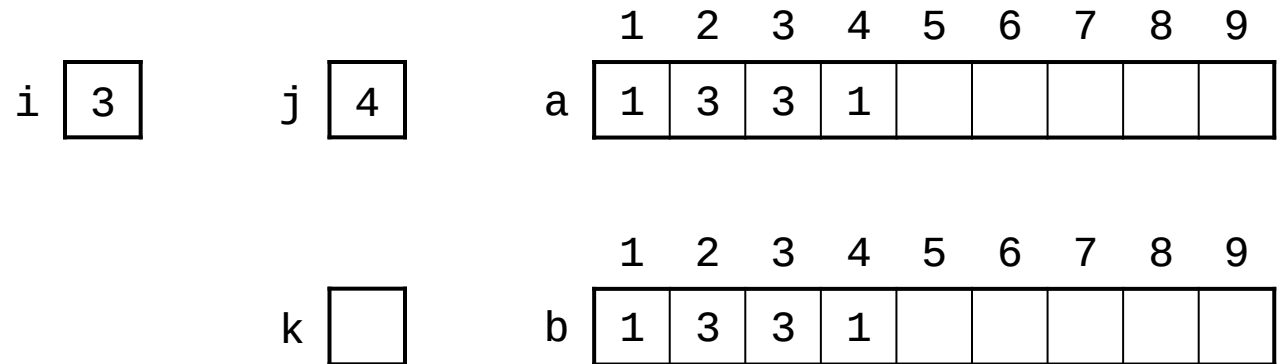


```

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{
    int level, a[100], b[100];
    cout << "How many levels? (1-13): ";
    cin >> level;
    a[1] = 1;
    for( int i = 1; i < level; i++ )
    {
        b[1] = a[1];
        for( int j = 2; j <= i; j++ )
            b[j] = a[j-1] + a[j];
        b[i+1] = 1;

        for( int k = 1; k <= i+1; k++ )
            a[k] = b[k];
    }
}

```



```

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            b[j] = a[j-1] + a[j];
        b[i+1] = 1;

        for( int k = 1; k <= i+1; k++ )
            a[k] = b[k];
    }
}
```

Pascal triangle - one arrays

	1	2	3	4	5	6	7	8	9
a	1								

				1					
			1		1				
		1		2		1			
	1		3		3		1		
1		4		6		4		1	

Pascal triangle - one arrays

	1	2	3	4	5	6	7	8	9
a	1	1							

				1					
			1		1				
		1		2		1			
	1		3		3		1		
1		4		6		4		1	

Pascal triangle - one arrays

	1	2	3	4	5	6	7	8	9
a	1	2	1						

				1					
			1		1				
		1		2		1			
	1		3		3		1		
1		4		6		4		1	

Pascal triangle - one arrays

	1	2	3	4	5	6	7	8	9
a	1	3	3	1					

				1					
			1		1				
		1		2		1			
	1		3		3		1		
1		4		6		4		1	